

AIR UNIT #12 — CARTOGRAPHER

ID:

obs-air-cartographer

Role: High-Precision Mapping / Survey Intelligence

Tier: Advanced

Status: Active

PHYSICAL & TECHNICAL

Length: 3.0 m

Width: 2.2 m

Height: 1.0 m

Mass: 26 kg

Power Source: High-capacity battery

Energy Capacity: 2500 Wh

Energy Consumption: 55 Wh/min

Endurance: 30–45 min

Payload Capacity: 2.0 kg

Modules: Multi-band LiDAR + optical mapping suite

Navigation System: GPS + Terrain Mapping AI

Structural Integrity: 135 HP

Armor Rating: 0.18

Heat Signature: Medium

Noise Signature: Medium

Top Speed: 20 m/s

Acceleration: Medium

Turning Radius: Tight

Climb Rate: Medium

Stability Rating: Very High (mapping precision)

AUTONOMY & AI BEHAVIOR

Behavior Profile: Mapping / Survey

Aggression: 0.0

Caution: 0.8

Persistence: 0.9

Flanking Likelihood: 0.0

Retreat Threshold: 0.7

Target Priority: Terrain features / structural points

Formation Discipline: High

Risk Tolerance: Low

Sensors

Visual Range: 1600 m

Thermal Range: 700 m

Acoustic Range: Low

Radar Range: None

LiDAR Range: 1800 m (multi-band)

Signal Detection Range: Medium

Communication

Mesh Range: 6 km

Relay Capacity: 0

Command Link Type: Mapping Grid

Network Priority: High

COMBAT & THREAT MODEL

Firepower Rating: 0

Accuracy: N/A

Stabilization: Very High

Weapon Mounts: None

Weapon Types: None

Suppression Value: 0

Tracking Strength: Medium

Disruption Strength: Low

Breach Capability: None

Countermeasure Rating: Low

Threat Modeling

Threat Level: 3

Engagement Distance: N/A

Preferred Terrain: Mapping corridors / open air

Preferred Targets: None

Vulnerability Profile: Ballistic / weather

SQUAD & COMMAND INTEGRATION

Squad Role: Mapping Intelligence

Squad Size Contribution: 1

Formation Behavior: Grid-pattern scanning

Intent Interpretation Rules: Maintain mapping altitude and coverage

Retreat Behavior: Medium

Breach Behavior: None

Flanking Behavior: None

Reinforcement Behavior: Supports recon and engineering squads

Command Hierarchy

Parent Command Unit: Insight / Command Core

Subordinate Units: None

Relay Nodes: None

Network Dependencies: Mapping Grid

VR OPERATOR MODE

Operator Seat Type: Drone View

Operator HUD: Multi-Band Mapping Overlay

Camera Feeds: Optical

Thermal View: Yes

LiDAR View: Yes (multi-band)

Sonar View: No

Tactical Overlay: Terrain mesh visualization

Operator Controls

Movement Controls: Full drone flight

Sensor Controls: LiDAR sweep / density adjustment / multi-band switching

Ability Controls: Scan, Map Terrain, Structural Sweep

Weapon Controls: None

Stabilization Controls: Auto-hover, mapping-lock

Operator Abilities

Multi-Band LiDAR Sweep

Terrain Mesh Generation

Structural Scan

Mark Terrain Features

Relay Mapping Data

ECONOMY & LOGISTICS

Build Cost: 210 Metal / 190 Electronics

Build Time: 65 sec

Factory Type: Drone Bay

Tech Requirements: Mapping Systems I

Maintenance Cost: Medium

Repair Time: 16 sec

Supply Consumption: Medium

Operational Cost: Moderate

Deployment Requirements: Standard drone pad

Transport Compatibility: All aerial racks

Launch Method: Vertical takeoff

BUILD REQUIREMENTS (PLAYER COLLECTION SYSTEM)

Common Components

14× **Medium Alloy Plates**

10× **Composite Frame Rods**

6× **Micro-Servo Actuators**

4× **Stabilizer Fins**

1× **Mapping Sensor Housing**

Electronics

14× **Circuit Bundles**

10× **Signal Processors**

6× **Drone Control Chips**

4× **LiDAR Processing Chips**

Power

1× **High-Capacity Battery Pack**

3× **Power Regulation Nodes**

Rare / Tradeable

1× **Multi-Band LiDAR Core**

1× **Terrain Mapping Microprocessor**

Optional Upgrades

Extended LiDAR Range Module

High-Density Scan Grid

Anti-Jamming Shielding

Silent Rotor Kit